

ERRATA — SIDE24 pre-peer-review manuscript (2026-08-05 edition)

Issued 2026-08-05, hours after delivery, on receipt of the K3 swarm work order's blocking-defect notice. The notice was independently verified and is CONFIRMED.

The defect

The window-saddle intensity upper bound used in the WP work (and reported in the manuscript as the “certified rigorous-class upper bound”) carries the factor

$$p_{\text{grad}}(0) \cdot \sqrt{E[\det(H)^2 \mid \text{grad}=0]} \cdot \min(P_{\text{type}}, P_{\text{window}}),$$

whereas the Cauchy–Schwarz chain gives

$$\begin{aligned} E[|\det H| \cdot 1_{\{\text{type} \cap \text{window}\}}] &\leq \sqrt{E[\det(H)^2]} \cdot \sqrt{P(\text{type} \cap \text{window})} \\ &\leq \sqrt{E[\det(H)^2]} \cdot \sqrt{\min(P_{\text{type}}, P_{\text{window}})}. \end{aligned}$$

Since $p \sim \sqrt{p}$ on $[0,1]$, the missing square root makes the reported quantity SMALLER — it is not a valid generic upper bound. Numerically, at the manuscript's witness point $(-0.04, -0.58)$: missing-sqrt value $1.31310\text{e-}4$ vs CS-valid $2.30247\text{e-}2$ (inflation $175.3\times$).

Manuscript items INVALIDATED (quarantined pending the K3 WP adjudication)

- §3.10.6, Proposition 3.23: the phrase “the certified rigorous-class upper bound (2.6) integrates to 1.30399×10^{-2} ” — NOT a valid upper bound. (The value itself is a correctly computed number; it is the upper-bound CLAIM that fails.)
- §3.10.6, Proposition 3.24: the “certified ub 1.047r^3 ” branch, the ub rung table's status as an UPPER bound, and the envelope (3.18) $\text{WP}(\text{r}) \leq 5.5\times 10^{-3}\text{r}^{1.6}$ as an upper-bound statement (it was fitted to the invalid surrogate). The two-sided TRUTH column stands as derived-on-grid ESTIMATES (not enclosures).
- §3.10.9, Proposition 3.26: the display using the WP ub branch (“certified WP ub 0.2666r^3 ”) — the constant $c = 0.91436$ at that branch is NOT established. The limit discussion $c \rightarrow 0.9144036$ is unaffected (it uses no WP bound).
- §3.10.6 W5b-equivalent claim “the printed channel total 0.213r^3 is falsified as an upper bound”: stands as derived-on-grid EVIDENCE (the estimate $4.76\text{e-}6$ exceeds it), but it is not yet a FORMAL refutation — a formal refutation requires a rigorous lower bound $> 3.328125\times 10^{-6}$, which is K3 workstream W3's target.
- §6.1/§6.2: all references to the items above (the grade map's WP row; C-3's envelope description).
- The THM-023 v1.1 draft cited in the manuscript was and remains a noncontrolling HOLD draft with placeholders; it is preserved unchanged and was never sealed.

Manuscript items UNAFFECTED (verified against the defect)

- LB-1's above-b saddle channel (§3.10.3): its factors $\rho_{\text{CS}}/\rho_{\text{sad}}$ carry the square root ($p_{\text{grad}} \cdot \sqrt{E[\det^2]} \cdot \sqrt{\min(\text{cant}, P)}$) — VALID; the six-rung E/tol chain stands.
- The two-sided quadrature ESTIMATE $I_{\text{true}} = 4.7602\times 10^{-6}$ (§3.10.6): a numerical estimate with displayed mesh/engine stability; never an enclosure claim. Its rigorous certification is K3 W3's

task.

- The conditional witness laws, $m(P) > b$ and $m(Q) > b$ (premise falsification, §3.10.6), the rim-band diagnosis, and the code-path origin analysis: no inequality used.
- Task 4c's Kantorovich enclosures (§3.10.4), DER-027a's variance channel (§3.10.7), -LOC's CONDITIONAL form (§3.10.8), SARD-G (§3.7), measurability (§3.8), the $\hat{-1/3}$ law (§3.9), the elder-rule/capture material (§3.2–3.3), Palm transfer (§3.4), collar theorems (§3.5), literature and status sections.

Disposition

The K3 swarm is executing the rigorous WP repair now (symbolic auditor, rigorous numerics, independent implementation, forensic provenance, uniform-in-r analysis, adversarial review). A corrected manuscript edition will be issued when the WP adjudication lands, with the four tiers (proved constant / limiting coefficient / measured anchor / direct numerical estimate) kept strictly distinct.

Addendum (red-team finding B2, same day)

The witness-point D2 values printed in the manuscript (and the numbers quoted in this errata) carried a fourth-moment display that drops mean terms (overcount exactly $2(\det \nu \nu^T)^2$; inflation $\sqrt{1.34488}$). Corrections: the defective-form witness value $1.31310\text{e-}4$ reads correctly $1.13229\text{e-}4$; the CS-valid value $2.30247\text{e-}2$ reads $1.98542\text{e-}2$; their ratio (the defect factor) is unchanged at 175.35; every quarantine and disposition in this errata stands. The manuscript's D2-dependent printed quantities inherit the same inflation (conservative direction); its two-sided estimate $4.7602\text{e-}6$ used direct quadrature and is unaffected. LB-1's above-b channel bounds are inflated but remain valid (margins absorb the factor).